The Attention Gap in Warming and Cooling Food Beliefs: A Two-Axis Bibliometric Map of Lay Belief Breadth versus Direct Research Attention for 76 Foods in Japan

Running title: Belief breadth is decoupled from research attention for warm/cool foods

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Abstract

Background: In Japan, the folk practice of *onkatsu* ("warming care") rests on a widely shared belief that foods warm or cool the body. Prior reviews establish that scientific support is heterogeneous rather than absent (Ormsby 2021; Zhou & Xu 2021), but none has quantified how *widely* a food is believed to warm or cool the body against how much *direct research attention* the claim has received.

Methods: We built a two-axis bibliometric map. Axis A (belief breadth) counts, per food, the independent Japanese lay-facing sources assigning a warming or cooling direction, coded from a frozen frame of 15 sources following García-Hernández et al. (2023); two-coder agreement was Cohen's $\kappa = 1.000$ over 649 rows. Axis B (research attention) counts PubMed hits for each food co-occurring with four thermal-effect terms (the "claimed-context" layer, L2), all queries logged. The analysis, planned before extraction, correlated the two axes for 76 core foods (Axis A ≥ 3 sources).

Results: Belief breadth and research attention were decoupled: Spearman $\rho = +0.057$ ($p = 0.62$; 95% CI −0.17 to +0.28) among core foods and $\rho = +0.128$ ($p = 0.18$) across all 110 foods — no monotonic association either way, with the confidence interval spanning zero. Seventeen of 76 core foods (22.4%) had zero claimed-context studies, and this fraction rose among the most widely believed (3 of 9 with ≥ 11 sources; 33.3%), led by carrot (13 sources, 0 studies), burdock (12, 0), and eggplant (11, 0). The warming zero rate (28.3%) exceeded the cooling rate (10.3%) numerically but not significantly (Fisher exact $p = 0.085$; odds ratio 3.41, 95% CI 0.88–13.26); warming and cooling foods did not differ in overall research attention either (Mann–Whitney $p = 0.54$). Coffee, popularly held to "cool," is dominated by caffeine, which *raises* core temperature (Peel 2025).

Conclusions: Lay belief breadth does not predict direct research attention for warming/cooling foods in Japan. The evidence base is fragmentary and unevenly attended — and, for at least one flagship food, oriented opposite to belief — rather than uniformly absent.

Keywords: attention gap; bibliometrics; folk nutrition; warming and cooling foods; thermoregulation; Japan

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Introduction

The idea that individual foods "warm" or "cool" the body is a durable feature of everyday health culture in Japan, where it underpins the popular practice of *onkatsu* (温活, "warming care") and lay reasoning about *hiesho* (冷え症, habitual cold sensitivity). Corporate wellness media, food manufacturers, and individual practitioners routinely publish lists that sort ginger, carrot, and burdock as "warming" and cucumber, tomato, and coffee as "cooling," and these attributions circulate widely enough to shape ordinary purchasing and seasonal eating advice. That the belief is culturally established, rather than niche, is visible both in the volume of lay-facing sources that maintain such lists and in the fact that Japanese epidemiology has taken the classification seriously: Nagata et al. (2017), analyzing the Takayama cohort (n = 28,356), applied four independent warm/cool food-classification lists to real dietary data. The premise of the belief is therefore neither obscure nor recent.

Crucially, the scientific literature on warm/cool food theory is not empty, and this study does not claim that it is. Ormsby (2021), in a scoping review of the nutritional evidence, characterized the support as "heterogeneous and of mixed quality" while identifying partial mechanistic correlates — heating foods associated with higher caloric density, sympathetic activation, and vasodilation, and cooling foods with higher water and fiber content and anti-inflammatory processes. Namiranian et al. (2021), a companion review in the same volume, surveyed the physiological basis of hot/cold theory across Persian medicine, traditional Chinese medicine, and Ayurveda. Zhou & Xu (2021) integrated candidate molecular mechanisms (for example, differential NF-κB and MAPK signaling) linking the cold/hot nature of foods to biological effects — while explicitly noting that research applying these mechanisms to foods is scarce relative to research on Chinese medicines. Any framing of the field as an "evidential desert" is thus untenable; the appropriate description is fragmentary, unintegrated, and unevenly attended.

At the level of individual foods, the available evidence is not only fragmentary but sometimes contradictory or opposite to belief. Ginger, the archetypal "warming" food, showed a significantly enhanced thermic effect of food in one small randomized crossover pilot (Mansour et al. 2012, n = 10 overweight men), yet a larger, double-blind crossover trial in a different population reported no such increase (Fagundes et al. 2021, n = 20 normal-weight women). Coffee, popularly classified as "cooling" (or "yin") in Japan, is dominated by caffeine, whose best-powered synthesis under thermal stress — a meta-analysis of k = 30 studies — found that caffeine significantly *raises* peak core temperature (Hedges' g = 0.44; Peel et al. 2025). The physiological direction most relevant to coffee's principal active compound therefore runs opposite to the folk attribution, even after allowing for the distinct measurement context (heat-exposure performance settings rather than everyday thermal sensation).

Methodologically, the practice of mapping a hot–cold belief system bibliometrically has direct precedent. García-Hernández et al. (2023) assembled and classified 101 academic publications spanning roughly a century to characterize Mexico's hot–cold system by research approach, depth, and conceptual domain. Nagata et al. (2017) demonstrated, within Japan, that the four extant warm/cool classification lists disagree with one another when applied to cohort data — an internal inconsistency that is itself a symptom of the field's uneven development. These works establish that belief systems of this kind can be quantified and that the classification schemes underlying them are neither unified nor exhaustively validated.

What remains unquantified is the *asymmetry* between the two facts above: that certain foods are very widely believed to warm or cool the body, and that the direct scientific attention paid to those specific claims varies enormously and does not obviously track belief. Ormsby (2021) asked whether mechanistic support exists but did not measure how widely each food is believed, nor compare research volume across foods; reviews of the adjacent clinical concept of cold hypersensitivity (Jin et al. 2025, 65 studies) consistently take traditional-medicine interventions, not lay food beliefs, as their subject. We therefore ask a different, descriptive question: for foods that Japanese lay sources widely classify as warming or cooling, is the breadth of that belief related to the amount of direct research attention the claim has received? We address it with a two-axis bibliometric map — belief breadth (Axis A) against claimed-context research attention (Axis B) — and report the resulting pattern.

Methods

Design

This is a descriptive, cross-sectional bibliometric study. It maps two independently constructed axes against each other for a fixed set of foods and reports their association; it does not estimate a causal effect of belief on research or vice versa. The analysis plan (the two-axis construction, the core subset defined by Axis A ≥ 3 sources, the Spearman correlation as the primary decoupling test, and the zero-attention and warm-versus-cool comparisons) was fixed before data extraction. Because the study was not registered on any protocol registry, we describe these choices as *planned* rather than pre-specified.

Axis A: belief breadth

Axis A quantifies how widely each food is attributed a warming or cooling direction in Japanese lay-facing sources, following the bibliographic-coding approach of García-Hernández et al. (2023). The sampling frame was frozen in advance: 15 sources verified to be reachable over HTTP, to present a concrete per-food warm/cool list, and to permit fetching per robots.txt. Nine corporate- or association-operated sources form the primary (Tier 1) frame; six individual-expert blogs form a Tier 2 set used only for sensitivity analysis. Sources sharing a single operator were counted once. Two frame corrections were made during fetching and before any coding of results — one source lacking a concrete per-food list was excluded, and one source's URL was corrected to its per-food list page — with both decisions driven by page content rather than by any result.

Each source page was fetched once and stored locally; the author then read the text and coded, for every food mentioned, its direction (warming, cooling, or neutral) with the supporting quotation and access date recorded. Coding rules were documented before coding: sources using traditional five-nature or yin–yang vocabulary were mapped to warming (温性/熱性/陽性), cooling (涼性/寒性/陰性), or neutral (平性). For each food, Axis A records the number of independent sources assigning a warming or cooling direction (belief breadth, `n_sources`), the majority direction, and a consensus ratio. To guard against a "subjective coding" critique, a second independent coding pass was performed over all coded rows; agreement was Cohen's $\kappa = 1.000$ across 649 rows and 15 sources (no disagreements), and every row was verified against the stored source text.

The analysis focused on the **core** foods — those reaching belief breadth ≥ 3 independent sources — as the set for which a food can be said to be "widely believed." Coding yielded 197 distinct foods across all sources; 79 reached ≥ 3 sources, of which 3 were composite category labels ("red meat and fish," "nuts," "spices") that cannot be turned into a single unambiguous search query and were excluded from Axis B, leaving **76 core foods**. A five-nature normative classification (as an external reference against which lay attribution could be compared) was planned but is deferred to future work because a citable primary source could not be secured within scope.

Axis B: research attention

Axis B quantifies direct research attention to each food's thermal claim using the NCBI PubMed E-utilities (no API key; queries rate-limited under 3 requests/s). All PubMed, CiNii, and OpenAlex queries were executed on 2026-08-06; because bibliometric counts grow over time, this snapshot date fixes the reported values. For each food we counted hits at three nested layers:

- **L1 (total):** the food term alone — total research volume, reported as a denominator only.
- **L2 (claimed context) — the primary Axis B metric:** the food term co-occurring with any of four thermal-effect terms — *thermogenesis*, *body temperature*, *peripheral circulation*, *thermoregulation* — i.e., studies that examine the food in the context of the warming/cooling claim.
- **L3 (design layer):** L2 further restricted to randomized controlled trials via the `[pt]` publication-type filter.

The four-term thermal vocabulary was frozen after a one-time validity check on real PubMed hits: adding human cold-sensitivity terms recovered no additional relevant studies for ginger, whereas adding plant "cold tolerance/sensitivity" terms injected agronomy noise (e.g., crop cold-tolerance studies for tomato), so the vocabulary was kept to the four thermoregulation terms. Japan-specific foods with weak English coverage were given minimal English synonyms (for example, natto → “fermented soybean”), and a small set of unambiguous binomials (for example, ginger → *Zingiber officinale*) were available for a MeSH sensitivity check; the inclusion criterion was held constant across all foods so that counts are comparable. Every query string and its raw JSON response were saved for auditability. OpenAlex (keyword-based, RCT filter unavailable) and the CiNii Research API (Japanese-language literature) were queried as auxiliary sources; the primary counts are from PubMed.

Analysis

The primary test of the study’s thesis is the Spearman rank correlation between belief breadth (`n_sources`) and claimed-context research attention (L2 hits) among the 76 core foods, with the same correlation across all counted foods reported alongside. Spearman (rather than Pearson) correlation was chosen because the L2 counts are heavily right-skewed and contain many ties at zero, making a rank-based measure the appropriate summary. Research attention was plotted as log10(L2 + 1) so that zero-study foods appear at the baseline rather than being dropped by a logarithmic scale. We report the zero-attention rate (share of core foods with 0 L2 studies) overall and among the most widely believed foods, and compared research attention between lay-warming and lay-cooling foods with the Mann–Whitney U test, with a Fisher exact test on the warm-versus-cool zero-attention contrast. All tests were two-sided at $\alpha = 0.05$. The core Spearman correlation is the single planned primary test; the all-foods correlation, the zero-rate comparisons, and the warm-versus-cool tests are reported descriptively without correction for multiple comparisons, and are interpreted accordingly. Ninety-five-percent confidence intervals are reported for the correlations (Fisher z-transformation with the Bonett–Wright variance for rank correlations) and for the warm-versus-cool odds ratio (Woolf’s logit method). Analyses used Python 3.14.3 with pandas 3.0.1, scipy 1.17.1, numpy 2.4.2, and matplotlib 3.10.8; the pipeline is covered by 48 unit tests (all passing), and every statistic reported here is recomputed from the pipeline output by `src/verify_stats.py`. Figure labels are in English (romanized food keys) to avoid CJK-glyph problems in the PDF build (weasyprint 68.1).

Coverage note

PubMed does not index J-STAGE/CiNii (Japanese), KMbase/OASIS (Korean traditional medicine), or CNKI (Chinese TCM). We queried CiNii for Japanese-language counts so that a zero or low claimed-context count reflects a genuine gap rather than a single-database artifact; the primary axis remains the logged PubMed L2 count, and the coverage limitation is carried into the Limitations.

Results

Sample

Axis A coding covered 197 distinct foods; 76 met the core criterion (belief breadth ≥ 3 independent sources) after excluding 3 composite labels. Among the 76 core foods, 46 were lay-classified as warming, 29 as cooling, and 1 as contested. Axis B returned PubMed counts for all core foods, with 110 foods counted in total once the sensitivity set (belief breadth = 2) was included (Table 1).

Belief breadth and research attention are decoupled

Belief breadth showed no association with claimed-context research attention. Among the 76 core foods, the Spearman correlation between `n_sources` and L2 hits was **$\rho = +0.057$ ($p = 0.62$; 95% CI −0.17 to +0.28)**; across all 110 counted foods it was **$\rho = +0.128$ ($p = 0.18$; 95% CI −0.06 to +0.31)** (Table 2). Both confidence intervals span zero and exclude even a moderate association in either direction. Neither is a monotonic relationship in either direction: the data are consistent with belief breadth and research attention being decoupled, and specifically do *not* support the intuition that more widely believed foods are better studied — nor its converse. The scatter of the two axes (Figure 1, all foods; Figure 2, core only) shows the two dimensions varying largely independently, with a persistent cluster of widely believed foods along the zero-attention baseline.

A concentration of belief with zero direct attention

Seventeen of the 76 core foods (**22.4%**) had zero claimed-context (L2) studies despite being classified as warming or cooling by at least three independent sources. This zero-attention fraction did not decline among the most widely believed foods; it rose: among the 9 core foods believed by ≥ 11 sources, 3 (**33.3%**) had zero L2 studies, and among the 18 foods believed by ≥ 9 sources, 5 (27.8%). The widely believed, entirely unstudied foods were led by carrot (13 sources, 0 studies), burdock (12, 0), and eggplant (11, 0); the full list of 17 appears in Table 3.

The research attention that does exist is highly skewed. Across the 76 core foods, the median L2 count was 3.5 and 48 of 76 (63%) had 5 or fewer claimed-context studies, while the mean of 31.0 was pulled upward by 7 foods with ≥ 100 L2 studies (e.g., milk, chicken, egg — foods whose large counts reflect general nutrition research rather than thermal-claim testing). Only 25 of 76 core foods (33%) had even one claimed-context randomized controlled trial (L3 ≥ 1).

Warming versus cooling foods

Warming foods had a numerically higher zero-attention rate (13/46, 28.3%) than cooling foods (3/29, 10.3%), but the difference did not reach significance (Fisher exact $p = 0.085$ on the zero-attention contrast; odds ratio 3.41, 95% CI 0.88–13.26). Overall research attention also did not differ between the two (Mann–Whitney $U = 610.5$, $p = 0.54$ on L2 counts; median L2 warming = 2.5, cooling = 4.0). We therefore do not claim a warm/cool asymmetry in research attention.

Flagship foods: attention present but thin, skewed, or reversed

Three foods illustrate that "attention present" does not resolve the gap (Table 4). **Coffee** (lay-cooling, 6 sources) had 23,286 total PubMed hits (L1) but only 24 in the claimed thermal context (L2; 0.1% of L1) and 4 claimed-context RCTs — and the best-powered relevant synthesis (Peel et al. 2025, k = 30) reports that caffeine *raises* core temperature (g = 0.44), opposite to the "cooling" belief. **Ginger** (lay-warming, 11 sources) had 5,730 L1 hits, 35 L2 hits, and 7 claimed-context RCTs, but the direct thermic-effect evidence is contradictory (positive in Mansour 2012, null in Fagundes 2021). **Green tea** (lay-cooling, 5 sources) had 15,476 L1 hits but 62 L2 hits (0.4%) with 11 RCTs — the most-attended claimed context in the core set, yet still a small fraction of the food's overall research volume. In each case, large total research volume coexists with thin, skewed, or belief-reversed attention to the specific thermal claim.

Discussion

Across 76 foods that Japanese lay sources widely classify as warming or cooling, the breadth of belief was decoupled from direct research attention: the rank correlation between the two axes was essentially zero ($\rho = +0.057$, $p = 0.62$), and neither a positive nor a negative monotonic relationship was present. The clearest feature of the map is not a correlation but a concentration — nearly a quarter of widely believed core foods, and a third of the most widely believed, have received no direct study of the thermal claim at all. This pattern is consistent with an *attention gap*: what a culture finds worth believing about food and body temperature, and what science has directly examined, are set by largely independent processes.

The finding refines rather than contradicts the prior reviews. Ormsby (2021), Namiranian et al. (2021), and Zhou & Xu (2021) each establish that the field holds partial, mechanistically suggestive evidence, so "no evidence" is the wrong description; our contribution is orthogonal — we show that whatever evidence exists is distributed with no relation to how widely each food is believed, and is absent entirely for many of the most-believed foods. Zhou & Xu's own observation that food-level research is scarce relative to Chinese-medicine research is echoed quantitatively here: the majority of core foods have five or fewer claimed-context studies, and only a third have a single relevant RCT. García-Hernández et al. (2023) showed that a hot–cold belief system can be bibliometrically mapped and found domain-specific knowledge gaps in Mexico; our two-axis design extends that approach from documenting that a system exists to quantifying, food by food, the mismatch between belief and attention. Nagata et al. (2017) found that Japan's four warm/cool classification lists disagree when applied to cohort data; that internal disunity and the attention gap we report are complementary symptoms of a field that has been culturally elaborated faster than it has been empirically consolidated.

The flagship foods sharpen the point. Coffee is instructive because it is not understudied in general — it carries tens of thousands of publications — yet only a fraction of a percent of that literature addresses the thermal claim, and the best-powered relevant synthesis points in the opposite direction to the "cooling" belief (Peel et al. 2025). This is the strongest single illustration that abundant research volume does not imply that a specific lay claim has been examined, let alone confirmed. Ginger shows the complementary problem: the claimed context has been examined, but the direct evidence is thin and contradictory (Mansour 2012 positive; Fagundes 2021 null). Both cases are consistent with the general pattern that belief breadth and direct attention are decoupled.

An important interpretive caution applies to caffeine. The evidence that caffeine raises core temperature comes from heat-exposure performance settings (sport, military, and industrial contexts) rather than from studies of everyday subjective warm/cool sensation, and physiological core-temperature change is a distinct measure from felt thermal comfort. We therefore treat the coffee reversal as a well-powered example of belief-incongruent physiological evidence, not as a refutation of the subjective experience the belief describes. This context distinction — held constant across foods so as not to bias the counts — is exactly the kind of nuance that the attention gap obscures when a claim is repeated widely but examined rarely.

Because this is a descriptive, ecological mapping rather than a designed test of an effect, we frame the results as consistent with an attention gap rather than as demonstrating a cause. The value of the map is diagnostic: it identifies, with logged and reproducible counts, which widely believed foods have received no direct scrutiny — carrot, burdock, eggplant, and the fourteen others in Table 3 — and thereby marks concrete targets for the direct-verification studies that the field currently lacks.

Limitations

First — Belief breadth is a source-count proxy, not a measure of conviction. Axis A counts how many independent lay sources assign a food a warming/cooling direction; it does not weight by readership, and breadth of appearance is not the same as depth of public conviction. The frame is 15 Japanese lay-facing web sources, frozen in advance and dominated by corporate/association wellness media; a different frame (e.g., print encyclopedias of *yakuzen*, or reader surveys) could shift individual foods' breadth values. The Tier 1/Tier 2 split showed the core pattern to be stable: restricting the frame to the nine Tier 1 (corporate/association) sources left 51 core foods with a near-identical null correlation (Spearman $\rho = +0.047$, $p = 0.75$, 95% CI −0.23 to +0.32) and a comparable zero-attention rate (9/51, 17.6%), against $\rho = +0.057$ and 17/76 (22.4%) for the full Tier 1+2 frame.

Second — Research attention is a keyword-count proxy. The L2 metric counts co-occurrence of a food term with four thermoregulation terms and cannot judge study quality, relevance, or whether a hit truly tests the thermal claim; a zero count means "no indexed study matched the claimed-context query," not "no relevant work could ever exist." The four-term vocabulary was validated once against real hits but may still miss studies phrased in other terms, and PubMed's English-language emphasis under-counts Japanese and other-language work — partially mitigated by the auxiliary CiNii query but not eliminated.

Third — Database coverage is incomplete. PubMed does not index J-STAGE/CiNii, KMbase/OASIS, or CNKI, so the counts under-represent Japanese, Korean traditional-medicine, and Chinese TCM literatures. We queried CiNii to check the gap for Japanese-language work, but a comprehensive multilingual synthesis was out of scope.

Fourth — Direction and classification are simplified. Each food's lay direction is a majority vote across sources, collapsing genuinely contested foods (one core food was contested) and neutral attributions; a normative five-nature classification, against which lay attribution could be benchmarked, was planned but deferred for lack of a citable primary source within scope.

Fifth — Cross-sectional snapshot. Both axes are measured at one time; lay sources are updated and PubMed counts grow, so the specific values will drift, although the structural decoupling is unlikely to reverse on the timescale of source updates.

Sixth — The caffeine context differs from the belief. The evidence that caffeine raises core temperature derives from heat-exposure performance settings, not everyday thermal sensation, so the coffee "reversal" should be read as belief-incongruent physiological evidence in a specific context, not as a general disproof of the folk claim.

Seventh — Descriptive and unregistered. The design is ecological and correlational; it identifies an association pattern (decoupling), not a causal mechanism, and the analysis, though planned before extraction, was not registered on a public protocol registry.

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Ethical considerations

This study used exclusively publicly available information: aggregate bibliographic counts from public literature databases (PubMed, CiNii, OpenAlex) and publicly published lay-media web pages. No individual-level, health, or personally identifying data were accessed, and no human subjects were recruited or contacted. Because the analysis is restricted to already-public aggregate bibliographic data and public web content, it does not meet the standard threshold for human-subjects research under either domestic (Japanese Ethical Guidelines for Medical and Biological Research Involving Human Subjects, 2021 revision, for research using publicly available information) or international (e.g., 45 CFR §46.102 for non-human-subjects data) frameworks; accordingly, no institutional review board approval or waiver was sought or required. The study was conducted in accordance with the principles of the Declaration of Helsinki where applicable to research using aggregate, publicly available data.

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Claude Opus 4.8 (Anthropic) was used to assist with code drafting, debugging, literature-search support, and manuscript editing. All data extraction, statistical outputs, and numerical results were independently checked by the author by re-executing the analysis code (Python 3.14.3; unit tests n = 48, all passing) and by recomputing every reported statistic directly from the pipeline output before writing. The author is solely responsible for the accuracy of the numerical results, the choice of specifications, the interpretation, and the conclusions. All references were verified against PubMed, CrossRef, and publisher records; DOIs and PMIDs were confirmed against primary source pages, and one cited work (Jin et al. 2025) is not indexed in PubMed and is cited by DOI.

Author Contributions (Contributor Roles Taxonomy, CRediT)

Mizuki Shirai: Conceptualization, Methodology, Investigation, Data Curation, Formal Analysis, Software, Writing — Original Draft, Writing — Review & Editing, Visualization, Project Administration.

Conflict of Interest

The author declares no conflicts of interest per International Committee of Medical Journal Editors (ICMJE) guidelines.

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Data Availability

All input data used in this study are publicly available. Axis B counts derive from the NCBI PubMed E-utilities (https://www.ncbi.nlm.nih.gov/books/NBK25501/), the CiNii Research API (https://support.nii.ac.jp/en/cinii/api/api_outline), and OpenAlex (https://openalex.org/); every query string and raw response is stored in the analysis repository. Axis A source pages are the 15 publicly published lay-media URLs listed in the repository's `data/sources.csv`, each with access date and supporting quotation.

Analysis code (Python 3.14.3, pandas 3.0.1, scipy 1.17.1, numpy 2.4.2, matplotlib 3.10.8, weasyprint 68.1) — including the source-fetching and text-extraction scripts, the claim-coding and Axis A aggregation modules, the Axis B query builders and PubMed/CiNii/OpenAlex collectors, the query logs, the scatter-plot pipeline, and the statistical verification script (`src/verify_stats.py`) that recomputes every number reported here — is publicly available at https://github.com/rehabilitation-collaboration/warming-cooling-foods (MIT License). All numerical results reported in this manuscript are reproducible from the analysis code at repository commit `687c32901d9e5a21427e947914afc4232ed71367`, recorded as the immutable reproducibility anchor at first posting.

Tables

Table 1. Sample composition

Set	Definition	n foods	Warming	Cooling	Contested/neutral
All coded	≥ 1 lay source	197	—	—	—
Sensitivity + core (counted)	belief breadth ≥ 2	110	—	—	—
Core (primary)	**belief breadth ≥ 3**	**76**	**46**	**29**	**1**

Coding covered 197 distinct foods across 15 sources (Tier 1: 9 corporate/association; Tier 2: 6 individual), 649 rows, two-coder Cohen's κ = 1.000. Of 79 foods reaching belief breadth ≥ 3, three composite labels ("red meat and fish," "nuts," "spices") were excluded from Axis B as non-queryable, leaving 76 core foods.

Table 2. Primary statistics: belief breadth versus research attention

Test	Set	Statistic	95% CI	p
Spearman ρ (n_sources vs L2)	core (n = 76)	+0.057	−0.17 to +0.28	0.62
Spearman ρ (n_sources vs L2)	all (n = 110)	+0.128	−0.06 to +0.31	0.18
Zero-attention rate (L2 = 0)	core	17/76 = 22.4%	—	—
Zero-attention rate, belief ≥ 11	core subset	3/9 = 33.3%	—	—
Zero-attention rate, belief ≥ 9	core subset	5/18 = 27.8%	—	—
Mann–Whitney U (warming vs cooling, L2)	core	U = 610.5	—	0.54
Fisher exact (warm vs cool zero-attention)	core	OR = 3.41	0.88 to 13.26	0.085

Spearman 95% CIs via Fisher z-transformation with the Bonett–Wright variance for rank correlations; the odds-ratio CI via Woolf's logit method. Median claimed-context (L2) hits among core foods = 3.5; 48/76 (63%) had ≤ 5 L2 hits; mean = 31.0 (pulled up by 7 foods with ≥ 100 L2 hits). 25/76 core foods (33%) had ≥ 1 claimed-context RCT (L3 ≥ 1). Warming median L2 = 2.5, cooling median L2 = 4.0.

Table 3. The seventeen widely believed core foods with zero claimed-context (L2) studies, by belief breadth

Food	Lay direction	Belief breadth (sources)	L2 studies
carrot	warming	13	0
burdock	warming	12	0
eggplant	cooling	11	0
green onion	warming	9	0
lotus root	warming	9	0
daikon	cooling	8	0
brown rice	warming	7	0
hojicha	warming	6	0
soy sauce	warming	5	0
nira (garlic chives)	warming	4	0
sansho (Japanese pepper)	warming	4	0
umeboshi	warming	4	0
amazake	warming	3	0
horse mackerel	warming	3	0
nori	cooling	3	0
nukazuke	warming	3	0
komatsuna	contested	3	0

Table 4. Flagship foods: total volume versus claimed-context attention

Food	Lay direction	Belief breadth	L1 total	L2 claimed context	L2/ L1	L3 RCTs	Direct evidence
coffee	cooling	6	23,286	24	0.1%	4	Caffeine *raises* core temperature (Peel 2025, k = 30, g = 0.44) — opposite to belief
ginger	warming	11	5,730	35	0.6%	7	Thermic effect positive (Mansour 2012, n = 10) but null in larger trial (Fagundes 2021, n = 20)
green tea	cooling	5	15,476	62	0.4%	11	Most-attended claimed context in the core set, yet a small fraction of total volume

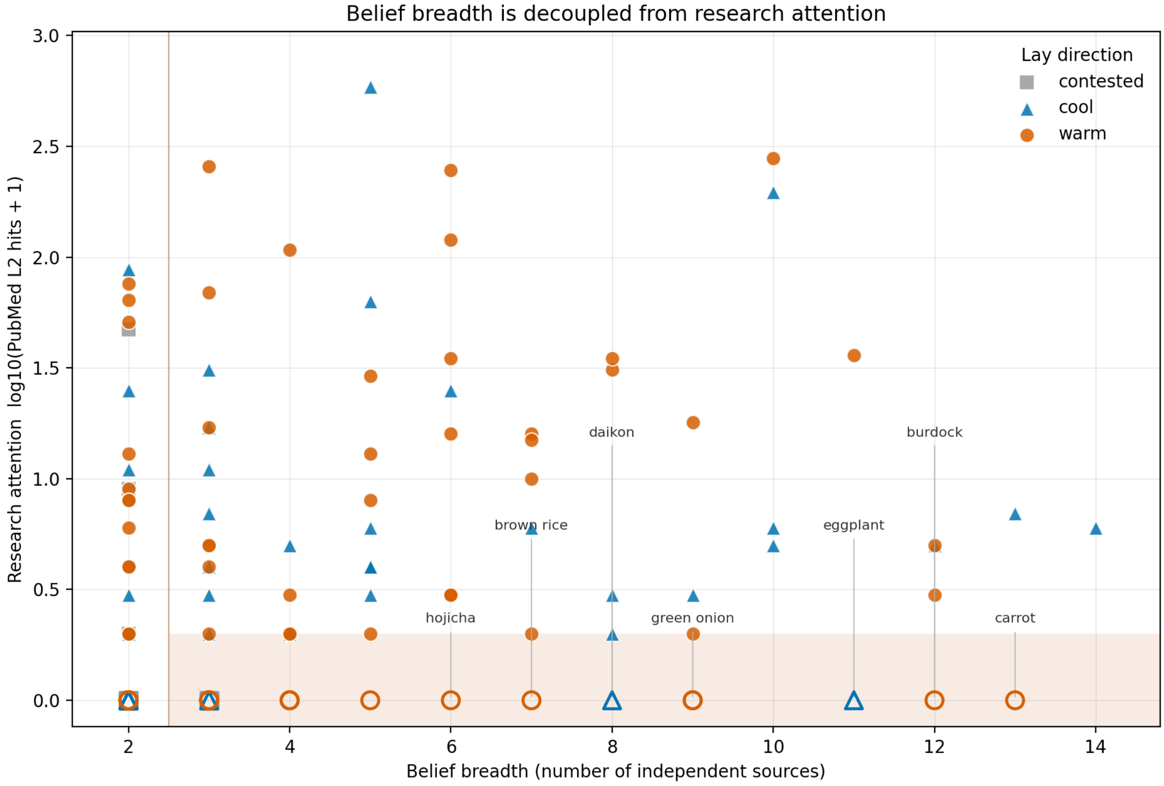

Figure 1. The attention gap across all counted foods (n = 110). Belief breadth (Axis A, x = number of independent lay sources assigning a warming/cooling direction) plotted against research attention (Axis B, y = log10(PubMed claimed-context L2 hits + 1)). Lay direction is encoded by both colour and marker shape (warming = vermillion circle, cooling = blue triangle, contested/neutral = grey square/diamond) so that identity never rests on colour alone. Foods with zero claimed-context studies are drawn with hollow markers on the y = 0 baseline; the shaded bottom-right region marks widely believed (n_sources ≥ 3) foods with little or no research attention. A selection of the widest-belief, zero-attention foods is directly labelled. Okabe–Ito colour-blind-safe palette.

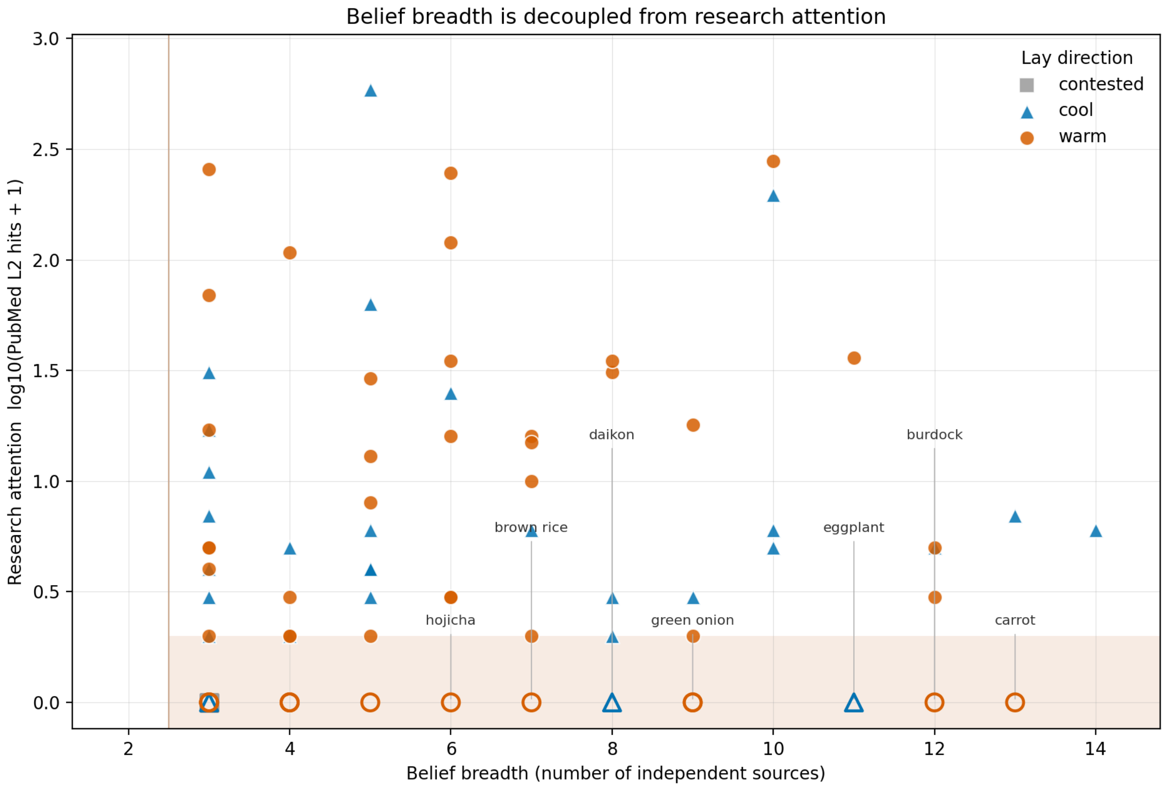

Figure 2. The attention gap among core foods (n = 76; belief breadth ≥ 3 sources). As Figure 1, restricted to the core set for which a food can be said to be widely believed. The persistence of the bottom-right cluster (17 foods at y = 0) illustrates the concentration of widely believed but directly unstudied foods; the near-zero Spearman correlation (ρ = +0.057, p = 0.62; 95% CI −0.17 to +0.28) is visible as the absence of any upward or downward trend.